NITHYA AJAYAN

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EDUCATION

Year	Degree	Institute	CGPA/%
2023 – present	B.Tech (Engineering Physics)	Delhi Technological University (DTU), New Delhi	8.39 CGPA
2023	12th CBSE	Indian School Sohar, Oman	95.4%
2021	10th CBSE	Indian School Sohar, Oman	95.8%

SUMMARY

Engineering Physics undergraduate with hands-on experience across the full satellite development lifecycle — from embedded systems and sensor integration to onboard telemetry and transient localisation algorithms. Contributed to the Daksha dual-satellite mission at Raman Research Institute; led miniature satellite projects as Captain of CanSat DTU. Seeking research or engineering roles in space instrumentation, satellite systems, and high-energy astrophysics.

INTERNSHIPS AND WORK EXPERIENCE

Daksha Mission, Raman Research Institute, Bangalore *May 2025 – Jul 2025*

- Developed and validated two independent transient localisation algorithms for the Daksha dual-satellite mission, using simulated detector responses across isotropically distributed sky sources.
- Modelled photon count distributions using Band function spectral parameters with realistic detector efficiency and background; recovered source directions with narrow residual spreads closely matching injected values.
- Benchmarked accuracy vs. computational cost trade-offs between the methods.

Embedded Systems Engineer Intern, India Space Lab, Delhi *Sep 2024 – Present*

- Architected and integrated the embedded systems of three miniature satellites, including sensor suites, power subsystems, and a recovery system.
- Established and validated long-range radio telemetry links for real-time onboard data downlink; conducted rigorous end-to-end testing of the acquisition and transmission pipeline.

Research Intern, RF & Antenna Systems, Pandit Deendayal Energy University, Virtual *Jan 2025 – Apr 2025*

- Designed the front-end of a 5G millimetre-wave antenna using Ansys HFSS, EM simulation and antenna characterisation.
- Studied broadband antenna architectures and fabrication constraints relevant to satellite communication payloads.

Embedded Systems Engineer Intern, Aerialbots, Delhi *Sep 2024 – Nov 2024*

- Evaluated VTOL UAV airframe configurations and contributed to structural design.
- Integrated camera systems for in-flight environmental analysis.

Embedded Systems Engineer Intern, Astroverse Pvt. Ltd., Nainital *Feb 2024 – Mar 2024*

- Built a fully functional miniature satellite prototype; secured long-range radio communication between satellite and ground station for live telemetry.
- Conducted rigorous pre-launch testing of the telemetry pipeline under simulated operational conditions.

Robotics Engineer Intern, MyEquation, Virtual *Dec 2023 – Jan 2024*

- Developed CAD models and embedded control logic for a Boston Dynamics 'Spot'-inspired quadruped robot using SolidWorks and Arduino.
- Applied basic AIML for sensor-based pattern recognition and autonomous behaviour scripting.

PROJECTS

Exoplanet Vetting Using Neural Networks

- Built MLP Classifier and XGBoost models to distinguish confirmed exoplanets from false positives across Kepler, K2, and TESS mission datasets; tested multiple architectures and achieved 92% classification accuracy.
- Performed feature engineering on photometric light curves and transit parameters using `astropy` and `scikit-learn`.

Solar Flare Classification

- Developed a CNN-hybrid model to classify solar flares (A/B/C/M/X classes) using HMI solar magnetogram images from SDO.
- Applied transfer learning and data augmentation to address class imbalance across flare severity levels.

Miniature Satellites, CanSat DTU

- Designed and developed miniature satellites with integrated sensor suites, onboard data acquisition, and live telemetry systems.
- Managed cross-functional workflows across avionics, mechatronics, and software subsystems as team captain.

Drone Platform for AQI Monitoring

- Designed a UAV equipped with particulate and gas sensors for real-time Air Quality Index (AQI) mapping.
- Implemented onboard data acquisition and downlink pipeline; visualised pollutant spatial distributions over monitored areas.

POSITIONS OF RESPONSIBILITY

Captain, Team CanSat DTU, Space Tech Team of DTU

- Leading a 20+ member interdisciplinary team for model satellite missions; overseeing avionics, mechatronics, software, corporate, and media.

General Secretary, Cosmology Club DTU

- Organised sessions on cosmology and observational astronomy; coordinated talks with research scientists and astronomers.

Vice President, Deltech Engineering Physics Technical Hub

- Designed and conducted technical events for inter-college fests; facilitated group learning sessions on physics topics.

TECHNICAL SKILLS

- **Satellite & Embedded Systems:** CubeSat integration, sensor interfacing, long-range RF telemetry, Arduino, embedded C, SolidWorks, Ansys HFSS
- **Data Science & ML:** TensorFlow, PyTorch, scikit-learn, XGBoost, NumPy, Pandas, Matplotlib, Seaborn
- **Astrophysics Tools:** `astropy`, `healpy`, LIGO library, MATLAB, Jupyter, Spyder
- **Coursework:** Cosmology & Astrophysics, Quantum Physics, Communication Systems, Stochastic Processes, Probability & Statistics, AIML